

Dustin Lin

Software Developer - ML Platform

Reliable and results-driven senior software engineer with 4+ years of experience in solving complex technical challenges and delivering optimized solutions. Skilled in system architecture, clean coding, and cloud development, with expertise in **data structures, solution design, and strategic improvements**.

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WORK EXPERIENCE

Senior Software Developer - ML Platform ODAIA Intelligence Inc.

01/2024 - Present

- Orchestrated **migration from RDS to Snowflake** in a **monorepo** environment, restructuring and recoding the entire ML pipeline to improve scalability and achieve \$120K in annual cost savings.
- Engineered **S3 ingestion** into snowflake with **csv.gzip format** for the most optimized speed, reducing 1M rows ingestion time from 30 minutes (RDS) to 6 seconds in Snowflake, significantly enhancing data processing efficiency.
- Built a **FastAPI application** for efficient data retrieval from **Snowflake** via **stored procedures**, improving pipeline processing time by 30% and accelerating UI data access by consolidating historical data for each individual doctor.
- Migrated data models to **dbt** and **Dagster** and integrated **Snowflake** table with **Sigma**, reducing report load times by **80%** and infrastructure costs by **\$10k/month** by eliminating legacy Snowflake queries and manual CSV generation.
- Automated deployment of **views and stored procedures** across multi-tenant Snowflake databases using **Python CLI library, Typer** with **Bitbucket pipelines**, eliminating manual updates and ensuring consistent logic for legal compliance.

Software Developer - ML Platform ODAIA Intelligence Inc.

04/2023 - 01/2024

- Optimized **Maptual Platform** to handle large client data by implementing file pre-generation using **multiprocessing** and **AWS Step Functions with Terraform**, reducing task runtime by 50% and enabling the platform to meet client demands at scale.
- Developed **Maptual Labs**, an internal tool with model and output sanity checks built in **Streamlit**, reducing QA debugging time by 80% and improving overall testing efficiency.
- Built a FastAPI application** to streamline data access, improve reliability, and enhance security by **eliminating direct S3 and RDS queries**, enabling dynamic data retrieval and simplifying tracking without rerunning tasks after configuration changes.

Software Developer - ML Platform Cymax Group Technologies

03/2021 - 04/2023

- Listing-Error Analyzer:** Generated over **\$100k in savings** by developing an AI-driven analyzer with a **React/Python UI** to automatically detect discrepancies across marketplace listings..
- SKU-Reversal Application:** Accelerated marketing strategy execution by **architecting** a computer vision application for furniture recognition, deploying the full stack (UI to backend) on **Kubernetes**.
- Automated Leadtime Calculator for Amazon:** Maintained late shipping rate under 4% in Amazon and reduced costs from support by automating the lead time calculation process with a Machine Learning based algorithm, **deployed on Kubernetes**.

TECHNICAL SKILLS

Programming Languages

Python, C++, SQL, HTML, JavaScript, JQuery, Swift

Python Libraries

Pandas, Numpy, DuckDB, Multiprocessing, TensorFlow, OpenAI, XGboost, SKlearn, Dynaconf, Unittest/Pytest

Frameworks

FastAPI, Streamlit, Django, Ruby on Rails, React, Flask, Bootstrap

Tools

Terraform, Sigma, Docker, Bitbucket, Kubernetes, Jenkins, Grafana/Zabbix, Postman, Weights & Biases

Databases

Snowflake, RDS, PL/SQL, MSSQL, GraphDB, Postgres

Cloud Services

AWS, ECS/ECR, Step Functions, CloudWatch, S3, Azure, Azure Container Registry/Blob Storage/App service

Methodologies

Monorepo, MVC Architecture, Relational Database, Microservices, CI/CD Pipeline

CERTIFICATES

Designing Microsoft Azure Infrastructure Solutions (04/2023) [🔗](#)

Microsoft Certified: Azure Administrator Associate (01/2023) [🔗](#)

Microsoft Certified: Azure Fundamentals (12/2021) [🔗](#)

Structuring Machine Learning Projects (10/2021) [🔗](#)

Deep Neural Networks (06/2021) [🔗](#)

Neural Networks & Deep Learning (05/2021) [🔗](#)

EDUCATION

Systems Engineering, Bachelor of Applied Science

Simon Fraser University

09/2014 - 08/2020